

MARL-based Trust Evaluation for Zero-Trust Low-Altitude UAV Networks

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Abstract

Low-altitude UAV networks face significant security challenges due to their rapidly changing topology, intermittent links, and limited onboard resources, making them vulnerable to spoofing, tampering, and collusive attacks. When malicious nodes constitute a high proportion of the swarm, existing trust evaluation schemes often suffer from degraded detection accuracy and insufficient mission continuity. Although recent studies such as GDM-DTM, FUBA, and GALTrust have proposed various trust management mechanisms, these methods rely on static trust weighting, depend on centralized coordination, or struggle to maintain robust performance under high malicious-node ratios. To address these limitations, this paper proposes MTIM, a Multi-agent Reinforcement Learning-based Trust Evaluation Mechanism for zero-trust low-altitude UAV networks. MTIM constructs a hierarchical trust evaluation framework that integrates multi-dimensional trust evidence with adaptive learning and uncertainty-aware aggregation. Specifically, MTIM incorporates five complementary trust dimensions—behavioral, communication, energy, task-execution, and historical trust—and combines them with four core mechanisms: cross-factor consistency checking, decentralized trust learning, Bayesian global aggregation, and temporally decayed state updates. In this architecture, multiple Thinker Agents adaptively adjust trust-factor weighting and incentive decisions based on local observations, while an Inferential Conclusion Agent fuses local estimates into a globally consistent trust score with explicit uncertainty quantification. Simulation results show that MTIM achieves a task success rate of 90.0% under 30% malicious nodes and improves detection accuracy by up to 29.0% over the best individual trust branch, consistently outperforming all compared baseline methods. These results demonstrate that combining multi-factor evidence with MARL-based adaptation significantly improves robustness, stability, and mission-level decision quality in dynamic zero-trust UAV environments, even under high malicious-node ratios.

Keywords: UAV networks; trust management; multi-agent reinforcement learning; zero trust; low-altitude economy

Received:

Revised:

Accepted:

Published:

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1. Introduction

Low-altitude UAVs are becoming critical infrastructure for logistics, inspection, sensing, and emergency response, and their flexibility and rapid deployment make them a cornerstone of the emerging low-altitude economy [1]. However, the openness and mobility that enable these applications also greatly enlarge the attack surface of UAV networks. In

flying ad hoc networks (FANETs), three fundamental characteristics complicate security protection: highly dynamic topology, intermittent connectivity, and severely constrained onboard resources. These characteristics make malicious node infiltration, data tampering, spoofing, and collusion particularly difficult to detect and isolate [2].

Trust management has emerged as a core security primitive for UAV networks, enabling continuous assessment of node trustworthiness to detect malicious activities and mitigate attack impacts. However, existing trust management schemes still face three persistent limitations:

(1) Many existing methods rely on fixed rules or centralized coordination, which weakens resilience in infrastructure-scarce missions common in low-altitude UAV operations. (2) Recent approaches such as GDM-DTM, FUBA, and GALTrust either depend on manually designed trust weighting, lack adaptive mechanisms for dynamic environments, or are optimized for static IoT settings rather than highly dynamic UAV swarms. They often fail to effectively detect coordinated attacks that simultaneously affect communication quality, task execution, and energy behavior [8,10,23]. (3) Most methods lack adaptive mechanisms that can dynamically adjust trust strategies in response to changing attack patterns, leading to significant accuracy degradation when malicious nodes account for a high proportion of the swarm.

Zero trust offers a principled security perspective for this setting, built on the core principle of “never trust, always verify”, which requires continuous context-aware verification rather than static perimeter-based trust [15]. However, applying zero trust to UAV swarms requires more than just authentication. It also requires online trust estimation, uncertainty management, and adaptive response mechanisms that can operate effectively under mobility, partial observability, and limited resources. Meanwhile, multi-agent reinforcement learning (MARL) has shown strong potential for distributed control and optimization in UAV systems [16], but it has rarely been leveraged to jointly optimize trust evaluation and incentive design, where trust assessment itself becomes the optimization objective to shape long-term trustworthy behavior.

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This paper proposes MTIM, a Multi-agent Reinforcement Learning-based Trust Evaluation Mechanism for zero-trust low-altitude UAV networks. The key idea behind MTIM is to integrate multi-dimensional trust evidence collection, decentralized adaptive learning, and Bayesian uncertainty aggregation into a unified closed-loop framework: multi-factor trust assessment provides node credibility scores, the incentive mechanism rewards high-trust nodes and penalizes low-trust nodes, and MARL optimizes the incentive strategy to improve long-term network security. This creates a positive feedback loop that progressively drives the network toward a more trustworthy state. The main contributions of this paper are as follows:

- **A Unified MARL-based Framework for Trust Evaluation and Incentive Design:** Existing methods treat trust evaluation as a fixed scoring task rather than a closed-loop process that shapes node behavior. We propose a unified zero-trust framework that integrates multi-dimensional trust assessment with reinforcement learning-based

incentives. In this closed loop, multi-dimensional trust assessment provides the credibility score for each node, the incentive mechanism rewards nodes with high trust and penalizes nodes with low trust, and MARL optimizes the incentive strategy to improve long-term network security. This creates a positive feedback loop that progressively drives the network toward a more trustworthy state.

- **Multi-factor Trust Evaluation with Cross-Validation:** Single-factor or manually weighted models fail to detect coordinated attacks affecting multiple dimensions simultaneously. We design a comprehensive trust evaluation mechanism with five complementary dimensions—Behavioral Trust, Communication Trust, Energy Trust, Task Execution Trust, and Historical Trust—and a cross-factor consistency checking mechanism. This design enables robust detection of sophisticated attacks that manipulate only a subset of trust factors.
- **Decentralized Multi-Agent Trust Learning:** Static trust weighting cannot adapt to dynamically changing network conditions. We develop a decentralized MARL architecture where multiple Thinker Agents operate in parallel. Each agent is responsible for a subset of UAV nodes, locally adaptively adjusts trust-factor weights based on its observations, and learns a reward function that jointly balances detection accuracy, incentive effectiveness, temporal stability, and resource consumption. This distributed parallel learning approach improves scalability and adaptability, enabling robust trust differentiation even when malicious nodes account for a high proportion of the swarm.
- **Hierarchical Aggregation with Uncertainty Quantification and Dynamic Evolution:** Local multi-agent assessments inevitably involve bias and uncertainty. We introduce an Inferential Conclusion Agent that fuses local estimates from multiple Thinker Agents into globally consistent trust scores using Bayesian inference, with explicit quantification of estimation uncertainty. Combined with a temporal decay mechanism that enforces continuous verification, this approach ensures that trust values always reflect both historical behavior and recent observations, maintaining effective trust assessment in highly dynamic UAV swarms.

The remainder of this paper is organized as follows. Section 2 reviews related work on trust management in UAV networks and multi-agent reinforcement learning for network security. Section 3 presents the system model, including the multi-factor trust evaluation mechanism, the RL-based multi-agent framework, the reward function design, and the state update mechanism. Section 4 describes the experimental setup and presents the performance evaluation results. Section 5 concludes the paper and outlines future research directions.

2. Related Work

Trust evaluation for UAV networks has evolved from static reputation mechanisms to evidence-fusion, blockchain-enabled, and learning-driven designs. However, the combination of zero-trust security principles, distributed trust learning, and uncertainty-aware aggregation remains underdeveloped in low-altitude UAV networks. This section reviews prior work from three perspectives: trust management in UAV/FANET systems, zero-trust and continuous trust assessment, and MARL for UAV networking and security.

2.1. Trust Management in UAV and FANET Systems

Recent surveys show that trust management has become a core security primitive in FANETs, where dynamic topology, intermittent links, and resource limits make purely identity-based protection insufficient [1–5]. Existing schemes fall into three categories: behavior-centric trust analytics, distributed or blockchain-supported trust, and learning-enhanced trust evaluation.

Behavior-centric trust evaluation widely uses fuzzy and evidence-fusion methods to combine heterogeneous observations. Benfriha et al. proposed FUBA, which uses fuzzy logic to integrate indicators such as RSSI, packet delivery ratio, delay, battery level, and humidity to identify insider threats in FANETs [8]. Li et al. developed an adaptive multi-granularity trust management scheme for UAV visual sensor security under adversarial attacks, where coarse- and fine-grained evidence is fused to improve robustness against known and unknown threats [9]. Akram et al. introduced GALTrust, which combines generative adversarial learning with type-2 fuzzy logic to improve adaptability to emerging malicious behaviors in spatial crowdsourcing drone services [10]. More recent federated-security studies also move trust assessment toward collaborative anomaly detection, including FL-based intrusion detection for FANETs, multimodal swarm anomaly detection, and communication-efficient FL for resource-constrained UAV-IoT systems [11–13]. These studies significantly enrich trust evidence modeling, but most remain tightly coupled to specific sensing, communication, or crowdsourcing scenarios, and generally rely on hand-crafted fusion pipelines or fixed trust update logic.

A second line of work improves trust persistence and auditability through distributed storage or hierarchical trust exchange. Ge et al. developed a provenance-aware distributed trust model for resilient UAV networks, highlighting the value of peer-to-peer evidence tracing in hostile environments [7]. Zuo et al. proposed a hierarchical blockchain-based trust measurement method for drone clusters to address trust inheritance, cross-cluster re-evaluation, and historical trust verification under dynamic node entry and exit [6]. Akram et al. proposed TMIoT, which integrates digital twins, blockchain, and deep-learning-based intrusion detection to secure Open RAN-based Internet of Drone Things applications [17]. Closely related secure-FL frameworks have further explored malicious-node detection and trusted collaboration through blockchain-supported service provisioning, privacy-preserving IoT security, and hierarchical federated aggregation [18–21]. Related work on secure quantum federated learning in UAV-assisted wireless networks further indicates that trust-enhanced game incentives can improve participation quality and strengthen learning robustness under strategic behavior [22]. More recently, Hu et al. proposed GDM-DTM for malicious-node detection in low-altitude UAV networks by combining group decision making with dynamic trust management [23]. Although these schemes improve distributed credibility and malicious-node detection, most depend on manually designed trust weighting and update rules, limiting their adaptability to coordinated multi-factor attacks and rapidly changing mission conditions.

2.2. Zero-Trust and Continuous Trust Assessment

Zero trust shifts security from perimeter-based assumptions to continuous, resource-centric verification. The NIST zero-trust architecture defines this paradigm through the principle of “never trust, always verify” and replaces implicit perimeter-based trust with continuous, context-aware verification [15]. Building on this idea, Wang et al. proposed a zero-trust-driven collaborative intrusion detection framework for IoT that uses continuous trust assessment to support distributed security decisions [24]. In the UAV and 6G context, Al Ridhawi and Aloqaily integrated zero trust, digital twins, federated learning, and blockchain into a UAV-enabled 6G framework, highlighting the broader relevance of decentralized trust enforcement in highly dynamic aerial networks [25]. In the UAV authentication domain, Wang et al. introduced a context-aware trust evaluation model for malicious UAV detection during authentication, showing that trust can also be injected into the authentication stage rather than being treated purely as a post hoc reputation signal [14].

Zero trust is particularly relevant to low-altitude UAV swarms: node roles may change during flight, links may appear and disappear rapidly, and local observations may be incomplete or noisy. Trust decisions cannot be based solely on prior membership or static credentials, but must be continuously updated from current observations, device status, communication behavior, and mission context. Therefore, trust decisions should be dynamic, uncertainty-aware, and resource-conscious rather than static or binary.

These studies move trust evaluation closer to continuous verification. Nevertheless, the current zero-trust literature still focuses mainly on authentication, access control, or intrusion detection. It does not fully address how trust should evolve online in low-altitude UAV swarms when heterogeneous evidence from behavior, communication quality, task execution, energy state, and historical interactions must be fused under strict resource constraints. In particular, uncertainty-aware aggregation and decentralized incentive adaptation remain underexplored in zero-trust UAV settings.

2.3. MARL for UAV Networking and Security

MARL has recently become a prominent framework for decentralized decision making in UAV systems. A recent survey by Ekechi et al. shows that MARL has mainly been used for UAV control, formation, coverage, coordination, and cooperative optimization, rather than for trust management itself [16]. Representative examples include the MADDPG-based multi-UAV trajectory planning framework of Wang et al. for UAV-assisted mobile edge computing [26], the semi-distributed multi-agent federated reinforcement learning approach of Nie et al. for resource management in UAV-aided MEC systems [27], and the joint optimization of trajectory control, task offloading, and resource allocation in air-ground integrated networks by Alam and Moh [28]. Closely related learning-and-optimization studies in UAV-enabled federated settings have examined asynchronous scheduling, contract-based participant selection, hierarchical personalized learning, air-ground FL coordination, and energy-aware trajectory and resource management [29–34]. These works confirm that distributed learning is well suited for optimization under partial observability and dynamic coupling among UAVs. More recently, Cao et al. explored LLM-based multi-agent situation awareness for zero-trust space-air-ground integrated networks [35]. Their results further suggest that model selection in security-oriented multi-agent systems should balance capability and deployment cost; for specialized subtasks, smaller task-adapted models can be more practical than larger general-purpose models.

However, existing MARL applications in UAV networking remain largely performance driven, with objectives such as delay reduction, energy efficiency, coverage, or offloading efficiency. The reward design in these studies is rarely aligned with trustworthiness, collusion resistance, or long-horizon security incentives. MARL has been used to optimize how UAVs move, communicate, or allocate resources, but not how they collaboratively evaluate trust, calibrate trust-factor importance, and propagate trustworthy decisions under zero-trust assumptions.

Overall, prior work has made strong progress in trust modeling, distributed storage, and UAV-oriented MARL, but three key challenges remain largely unaddressed:

(1) **Accurate trust assessment under high malicious-node ratios:** Existing methods often suffer from significant accuracy degradation when malicious nodes account for a large proportion of the swarm, leading to poor mission continuity.

(2) **Adaptability to coordinated multi-factor attacks:** Most existing schemes rely on single-factor metrics or manually designed trust weighting, which cannot effectively detect sophisticated coordinated attacks that simultaneously affect multiple dimensions of node behavior.

(3) **Low-overhead continuous verification in dynamic environments:** Blockchain-based approaches introduce excessive coordination, storage, and consensus overhead that is prohibitive for resource-constrained UAVs, while non-blockchain methods struggle to adapt trust updates to rapidly changing topologies and mission conditions.

MTIM directly addresses these three challenges by combining multi-factor trust evaluation, decentralized MARL-based trust and incentive learning, Bayesian uncertainty-aware aggregation, and temporal trust decay into a unified zero-trust framework for low-altitude UAV networks.

3. System Model

Low-altitude UAV networks operate in dynamic environments characterized by rapidly changing topology, intermittent connectivity, limited onboard resources, and heterogeneous node capabilities. In such open and dynamic networks, UAV swarms are particularly vulnerable to spoofing, tampering, and collusive attacks. When malicious nodes constitute a high proportion of the swarm, sparse observations and intermittent communication make accurate trust assessment inherently uncertain.

If trust is inferred from only a single signal or updated too slowly, malicious behavior can propagate quickly across the swarm. This motivates a trust mechanism that continuously fuses heterogeneous evidence from multiple dimensions and adapts its decision policy as the network state evolves, maintaining accurate trust assessment even under high malicious-node ratios.

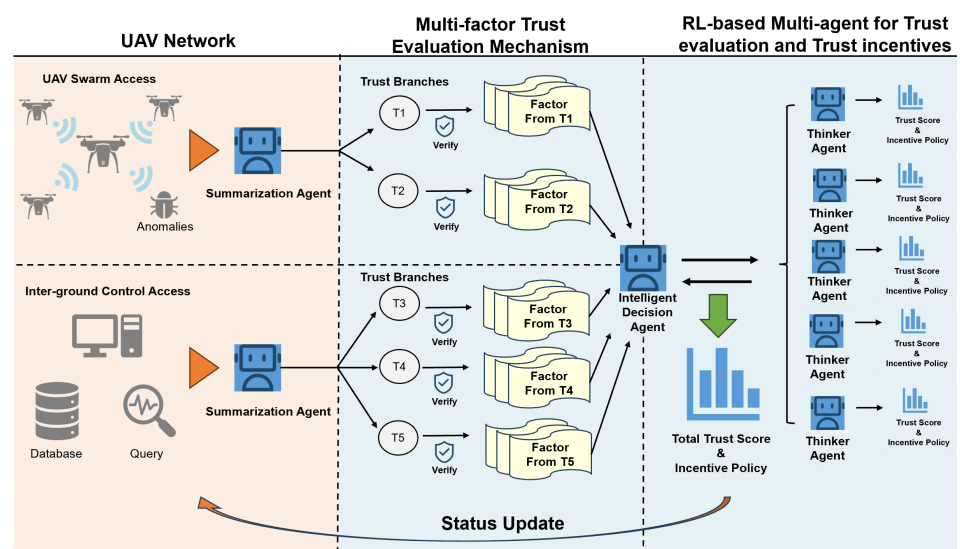


Figure 1. MARL-enabled Intelligent Trust Evaluation for Zero-Trust Low-Altitude UAV Networks framework.

The framework operates as follows. The UAV network environment provides multi-dimensional observations, including behavioral, communication, energy, task-execution, and historical trust signals. Cross-factor consistency checking detects coordinated attacks that manipulate only a subset of trust factors. Multiple Thinker Agents process these observations in parallel, each making local trust decisions for a subset of UAV nodes. The Inferential Conclusion Agent aggregates these local estimates via Bayesian inference to produce globally consistent trust scores with uncertainty quantification. Based on the trust evaluation, the incentive mechanism rewards trustworthy nodes and penalizes malicious ones, while MARL-based adaptation continuously optimizes the trust-factor weighting and incentive strategy. This closed-loop process embodies the zero-trust principle of continuous verification.

3.1. UAV Network Model

We model the low-altitude UAV network as a dynamic ad-hoc network consisting of N heterogeneous UAV nodes, denoted as $\mathcal{U} = \{U_1, U_2, \dots, U_N\}$. Each UAV node has limited communication range, computational resources, and energy capacity, and can dynamically join/leave the network based on mission requirements and environmental conditions. The network topology is represented by a time-varying graph $G(t) = (\mathcal{U}, \mathcal{E}(t))$, where $\mathcal{E}(t)$ denotes the set of communication links at time t . A link exists between U_i and U_j if their Euclidean distance $d_{ij}(t) \leq R_{comm}$, where R_{comm} is the maximum communication range (300 m as defined in Table 1).

3.2. Multi-factor Trust Evaluation Mechanism

In the proposed zero-trust architecture for low-altitude UAV networks, the multi-factor trust evaluation mechanism assesses UAV credibility from multiple complementary dimensions. It integrates heterogeneous trust evidence derived from direct observations and indirect references, enabling resilient trust assessment under dynamic and adversarial operating conditions.

3.2.1. Trust Factor Decomposition

To comprehensively capture different aspects of node behavior and detect sophisticated attacks that may only manifest in specific dimensions, we decompose trust evaluation into five complementary trust dimensions. This multi-factor design is motivated by the observation that a malicious node may behave normally in one dimension to evade detection while exhibiting abnormal patterns in another; by covering multiple dimensions, we increase the likelihood of detecting such coordinated attacks. Although including more factors introduces some computational overhead, each dimension captures a fundamentally different source of evidence, and the distributed MARL architecture handles this complexity efficiently.

As illustrated in Figure 1, the trust evaluation is decomposed into multiple trust branches, each corresponding to a distinct trust factor. Let the overall trust score of UAV node i at time t be denoted as $T_i(t) \in [0, 10]$, which is a function of five fundamental trust dimensions:

$$T_i(t) = 10 \cdot \sum_{m=1}^5 \omega_m(t) \cdot T_i^{(m)}(t) \quad (1)$$

where each trust branch $T_i^{(m)}(t) \in [0, 1]$ is defined as follows:

- **T1: Behavioral Trust** $T_i^{(1)}(t)$. This factor evaluates operational consistency, including adherence to flight paths, communication regularity, and response reliability:

$$T_i^{(1)}(t) = \frac{1}{N_b} \sum_{j=1}^{N_b} \mathbb{I}(a_{i,j}(t) = a_{i,j}^{expected}(t)) \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function, $a_{i,j}(t)$ represents the actual action in behavior category j , and $a_{i,j}^{expected}(t)$ is the expected behavior. This yields $T_i^{(1)}(t) \in [0, 1]$.

- **T2: Communication Trust** $T_i^{(2)}(t)$. This factor assesses link quality and stability through packet delivery ratio, signal strength, and latency:

$$T_i^{(2)}(t) = w_p \cdot PDR_i(t) + w_s \cdot \frac{RSSI_i(t)}{RSSI_{max}} + w_l \cdot \left(1 - \frac{L_i(t)}{L_{max}}\right) \quad (3)$$

where $PDR_i(t)$, $RSSI_i(t)/RSSI_{max}$, and $1 - L_i(t)/L_{max}$ are each normalized to $[0, 1]$, and w_p, w_s, w_l are weighting coefficients with $w_p + w_s + w_l = 1$. Thus $T_i^{(2)}(t) \in [0, 1]$.

- **T3: Energy Trust** $T_i^{(3)}(t)$. This factor monitors energy-consumption patterns and remaining battery life to detect anomalies indicative of malfunction or compromise:

$$T_i^{(3)}(t) = \frac{E_{remaining}(t)}{E_{initial}} \cdot \exp(-\alpha \cdot |C_i(t) - \bar{C}(t)|) \quad (4)$$

where $\frac{E_{remaining}(t)}{E_{initial}} \in [0, 1]$ and the exponential term is also bounded in $[0, 1]$, yielding $T_i^{(3)}(t) \in [0, 1]$.

- **T4: Task Execution Trust** $T_i^{(4)}(t)$. This factor measures the success rate and quality of task completion, such as sensing, data relaying, or payload delivery:

$$T_i^{(4)}(t) = \frac{1}{N_t} \sum_{k=1}^{N_t} \left(\frac{Q_{i,k}(t)}{Q_k^{max}} \cdot \mathbb{I}(task_success) \right) \quad (5)$$

where each term is normalized to $[0, 1]$, so $T_i^{(4)}(t) \in [0, 1]$.

- **T5: Historical Trust** $T_i^{(5)}(t)$. This factor incorporates long-term reputation scores and past behavior records to preserve temporal continuity in trust evaluation:

$$T_i^{(5)}(t) = \beta \cdot T_i^{(5)}(t-1) + (1 - \beta) \cdot T_i^{inst}(t) \quad (6)$$

where both the historical and instantaneous components are in $[0, 1]$, so $T_i^{(5)}(t) \in [0, 1]$.

3.2.2. Multi-factor Trust Aggregation

Each trust branch independently computes a sub-score $T_i^{(m)}(t) \in [0, 1]$ based on real-time data streams and historical logs. These sub-scores are then aggregated into a global trust score $T_i(t) \in [0, 10]$ through a weighted fusion mechanism:

$$T_i(t) = 10 \cdot \sum_{m=1}^5 \omega_m(t) \cdot T_i^{(m)}(t) \quad (7)$$

where $\omega_m(t)$ are dynamic weighting coefficients satisfying $\sum_{m=1}^5 \omega_m(t) = 1$, and the factor of 10 maps the aggregated score to the $[0, 10]$ range consistent with the classification thresholds $\theta_{high} = 7.5$ and $\theta_{low} = 3.0$. These weights are adaptively adjusted by the reinforcement learning agents introduced in the next section, allowing the system to emphasize more reliable trust factors under varying operational conditions.

3.2.3. Cross-Validation Mechanism

The multi-factor design ensures that no single point of failure or manipulation can compromise the overall trust decision. To enhance robustness, a **cross-validation mechanism** is implemented between trust branches. For any pair of trust factors m and n , the consistency metric is defined as:

$$\Gamma_{m,n}(t) = |T_i^{(m)}(t) - T_i^{(n)}(t)| \quad (8)$$

If $\Gamma_{m,n}(t) > \gamma_{threshold}$, a potential inconsistency is flagged, triggering deeper inspection. For instance, even if a UAV exhibits normal communication behavior ($T_i^{(2)}(t)$ high), but shows abnormal energy depletion ($T_i^{(3)}(t)$ low), the system can still flag it as potentially

compromised. This layered verification aligns with the **zero-trust principle** of "never trust, always verify," where trust is continuously evaluated and never assumed.

Furthermore, the framework supports **anomaly detection** through statistical analysis of trust factor deviations:

$$\mathcal{A}_i(t) = \mathbb{I} \left(\sum_{m=1}^5 \left(\frac{T_i^{(m)}(t) - \mu_m(t)}{\sigma_m(t)} \right)^2 > \chi_{\alpha,5}^2 \right) \quad (9)$$

where $\mu_m(t)$ and $\sigma_m(t)$ are the mean and standard deviation of trust factor m across the swarm, and $\chi_{\alpha,5}^2$ is the chi-squared threshold with 5 degrees of freedom at significance level α . This redundancy and diversity in trust evidence form the foundation for robust and adaptive trust management in UAV swarms, enabling the detection of sophisticated attacks that may manipulate multiple dimensions simultaneously.

3.3. RL-Based Multi-agent Trust Evaluation and Incentive Model

We formulate a multi-agent reinforcement learning (MARL) framework that jointly optimizes trust evaluation and incentive design for zero-trust UAV networks. Multiple Thinker Agents operate in parallel, each responsible for generating trust scores and incentive policies for the UAV nodes under its supervision.

3.3.1. System Model and Problem Formulation

We consider a UAV swarm consisting of N UAV nodes. The trust management system employs M Thinker Agents, where each agent $j \in \{1, 2, \dots, M\}$ is responsible for a subset of UAV nodes. The overall system can be modeled as a decentralized partially observable Markov decision process (Dec-POMDP):

$$\mathcal{G} = \langle \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle \quad (10)$$

where $\mathcal{S}$ is the global state space, $\mathcal{O}$ is the joint observation space, $\mathcal{A}$ is the joint action space, $\mathcal{P}$ is the state transition probability, $\mathcal{R}$ is the reward function, and $\gamma \in [0, 1]$ is the discount factor.

State Space: The state $s \in \mathcal{S}$ at time t is defined as the tuple of all UAV nodes' multi-factor trust scores:

$$s(t) = (\mathbf{T}_1(t), \mathbf{T}_2(t), \dots, \mathbf{T}_N(t)),$$

where $\mathbf{T}_k(t) = [T_k^{(1)}(t), T_k^{(2)}(t), T_k^{(3)}(t), T_k^{(4)}(t), T_k^{(5)}(t)]$ denotes the five-dimensional trust vector for node k , with each component in $[0, 1]$. The corresponding global trust score $\mathcal{T}_k(t) \in [0, 10]$ is computed by the ICA as described in Section 3.3.3.

Observation Space: Each Thinker Agent j observes a partial view of the global state, consisting only of the local trust sub-scores for the nodes in its supervised subset $\mathcal{N}_j$: $o_j(t) = (\mathbf{T}_k(t))_{k \in \mathcal{N}_j} \cup (c_{jk}(t))_{k \in \mathcal{N}_j}$, where $c_{jk}(t)$ is the confidence score assigned by agent j to its own assessment of node k .

Action Space: The action $a_j(t) \in \mathcal{A}_j$ for Thinker Agent j consists of two components: (1) the trust weight vector $\omega_j(t) = [\omega_j^{(1)}(t), \dots, \omega_j^{(5)}(t)]$ controlling how much emphasis to place on each trust factor, and (2) the incentive allocation vector $\pi_j(t) = [\pi_{jk}(t)]_{k \in \mathcal{N}_j}$ specifying the incentive level for each supervised node. The discrete action space for trust weights uses 11 quantization levels (0.0 to 1.0 in steps of 0.1), and incentive levels are quantized into 5 bins: $\{-2, -1, 0, 1, 2\}$ corresponding to penalty levels and reward levels.

Architecture: We adopt a Centralized Training with Decentralized Execution (CTDE) paradigm. During training, a global critic network has access to the full state $s(t)$ to

compute baselines for variance reduction, while each Thinker Agent's Q-network learns from its local observation $o_j(t)$ and the shared global reward. During execution, each Thinker Agent acts based solely on its local observation $o_j(t)$, ensuring scalability and practicality in deployed UAV networks.

State Transition: The state transition $\mathcal{P}$ is defined implicitly through the trust update equations (12)–(14) in Section 3.3.4, which govern how trust scores evolve based on observations, incentives, and temporal decay.

Reward Function: Defined in Section 3.3.3, the reward aligns local trust decisions with global network security objectives.

3.3.2. Thinker Agent Q-Network Architecture

Each Thinker Agent implements a deep Q-network (DQN) to learn an optimal policy for trust evaluation and incentive allocation. The Q-network takes the agent's observation $o_j(t)$ as input and outputs Q-values for all possible action combinations.

The Q-network architecture consists of an input layer, two hidden layers, a trust weight head, and an incentive head. The input layer maps the observation vector $o_j(t) \in \mathbb{R}^{5 \cdot |\mathcal{N}_j| + |\mathcal{N}_j|}$ to a hidden representation of size 128, followed by ReLU activation. The hidden layers are fully connected layers of size 256 and 128, with ReLU activations and layer normalization. The trust weight head outputs Q-values for each trust factor weight combination, using a factorized action head that independently outputs Q-values for the weight vector and the incentive vector before combining them. The incentive head outputs Q-values for each possible incentive action per supervised node.

The Q-value for a state-action pair (s, a) is computed as:

$$Q_j(s, a; \theta_j) = Q_j^\omega(s, \omega; \theta_j^\omega) + \lambda_{inc} \cdot Q_j^\pi(s, \pi; \theta_j^\pi) \quad (11)$$

where Q_j^ω and Q_j^π are separate value heads sharing the common feature extractor θ_j , and λ_{inc} balances trust evaluation and incentive objectives.

Each Thinker Agent updates its Q-network using the standard DQN update rule:

$$\theta_j \leftarrow \theta_j + \alpha \cdot [y_j - Q_j(s, a; \theta_j)] \cdot \nabla_{\theta_j} Q_j(s, a; \theta_j) \quad (12)$$

where $\alpha = 0.001$ is the learning rate, and the target $y_j = r_j + \gamma \cdot \max_{a'} Q'_j(s', a'; \theta_j^-)$ is computed using a target network Q'_j with parameters θ_j^- . The target network is updated every 100 steps. Experience replay uses a buffer of size 10,000, and mini-batches of size 32 are sampled uniformly at each update.

During training, each Thinker Agent uses an ϵ -greedy exploration policy with ϵ decaying linearly from 1.0 to 0.05 over the first 500 episodes, after which ϵ remains constant at 0.05 to ensure continued exploration while exploiting learned policies.

3.3.3. Reward Function Design for Thinker Agents and Global Trust Aggregation

In the proposed MARL framework, the effectiveness of trust evaluation and incentive mechanisms depends on reward functions that properly align local learning objectives with network-wide security goals. This subsection presents the reward structure, the aggregation mechanism implemented by the Inferential Conclusion Agent, and the state update dynamics that steer the UAV network toward higher trustworthiness.

To address the credit assignment problem inherent in multi-agent trust evaluation, we use a delayed reward mechanism where the ground-truth trust label for a node is determined by the consensus of *all* Thinker Agents' long-term observations, rather than relying on any single agent's instantaneous assessment. Specifically, we maintain a delayed

reward buffer where the ground-truth label for node k at time $t - \tau$ is determined at time t based on the majority vote of all agents' accumulated observations.

Each Thinker Agent j receives a composite reward signal $R_j(t)$ at each time step t , designed to capture multiple objectives: accurate trust assessment via delayed consensus, effective incentive allocation, and network-wide security enhancement. The reward function is formulated as:

$$R_j(t) = \lambda_1 R_j^{\text{consensus}}(t) + \lambda_2 R_j^{\text{inc}}(t) + \lambda_3 R_j^{\text{stab}}(t) - \lambda_4 R_j^{\text{cost}}(t) \quad (13)$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are weighting coefficients satisfying $\sum_{i=1}^4 \lambda_i = 1$, and each component is defined as follows:

Consensus-based Trust Accuracy Reward $R_j^{\text{consensus}}(t)$: To avoid circular dependency, we use a delayed external validation signal. The ground-truth label $\tilde{T}_k(t - \tau)$ for node k at time $t - \tau$ is determined at time t through a two-stage mechanism: (1) majority voting among all Thinker Agents' estimates at time $t - \tau$, and (2) confirmation by the ICA's global classification decision at time t , which incorporates new observations unavailable to individual agents. Concretely:

$$\begin{aligned} \tilde{T}_k(t - \tau) = & \text{median}\left(\left\{\hat{T}_{jk}(t - \tau) : j \in \mathcal{M}_k(t - \tau)\right\}\right) \cdot \mathbb{I}\left(C_k^{\text{ICA}}(t) = C_k^{\text{ICA}}(t - \tau)\right) \\ & + \mathcal{T}_k^{\text{ICA}}(t - \tau) \cdot \mathbb{I}\left(C_k^{\text{ICA}}(t) \neq C_k^{\text{ICA}}(t - \tau)\right) \end{aligned} \quad (14)$$

where $C_k^{\text{ICA}}(t)$ is the ICA's classification decision at time t (Trusted/Suspicious/Malicious), and $\mathcal{T}_k^{\text{ICA}}(t - \tau)$ is the corresponding ICA trust score. The indicator ensures the validation signal is anchored to the historical consensus unless new evidence explicitly overturns it. The delayed reward is then computed as:

$$R_j^{\text{consensus}}(t) = \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} \left[1 - \frac{|\hat{T}_{jk}(t - \tau) - \tilde{T}_k(t - \tau)|}{10} \right] \quad (15)$$

where $\tau = 5$ time steps is the delay buffer. This breaks the circular dependency because the validation signal $\tilde{T}_k(t - \tau)$ is determined by the broader system state at time t (including observations that agent j did not have access to at time $t - \tau$), not solely by agent j 's own assessment.

Incentive Effectiveness Reward $R_j^{\text{inc}}(t)$: This component evaluates whether the incentive action taken by agent j at time $t - \tau$ leads to improved ICA trust scores at time t :

$$\begin{aligned} R_j^{\text{inc}}(t) = & \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} \left[(\mathcal{T}_k(t) - \mathcal{T}_k(t - \tau)) \cdot \mathbb{I}(a_{jk}^{\text{inc}}(t - \tau) > 0) \right. \\ & \left. - \eta \cdot \mathbb{I}(\mathcal{T}_k(t) - \mathcal{T}_k(t - \tau) < 0 \wedge a_{jk}^{\text{inc}}(t - \tau) > 0) \right] \end{aligned} \quad (16)$$

where $\mathcal{T}_k(t)$ is the ICA's global trust score for node k , and $\eta = 0.5$ is a penalty factor for incentives that fail to improve trust. This formulation removes the circular dependency by using ICA-computed global trust scores (which incorporate external observations) rather than the agent's own estimates.

Temporal Stability Reward $R_j^{\text{stab}}(t)$: This encourages stable and consistent trust evaluation, penalizing erratic fluctuations:

$$R_j^{\text{stab}}(t) = 1 - \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} \frac{|\hat{T}_{jk}(t) - \hat{T}_{jk}(t - 1)|}{\max(10, |\hat{T}_{jk}(t - 1)|)} \quad (17)$$

where the denominator uses 10 as the maximum to normalize for the $[0, 10]$ trust scale. 454

Resource Cost Penalty $R_j^{cost}(t)$: This penalizes excessive resource consumption in trust evaluation and incentive execution: 455

$$R_j^{cost}(t) = \frac{1}{|\mathcal{N}_j|} \sum_{k \in \mathcal{N}_j} \left(\alpha_c \cdot c_{jk}^{eval}(t) + \beta_c \cdot c_{jk}^{inc}(t) \right) \quad (18) \quad 457$$

where $c_{jk}^{eval}(t)$ and $c_{jk}^{inc}(t)$ represent the computational and communication costs for evaluation and incentive actions, respectively, normalized to $[0, 1]$. 458

The Inferential Conclusion Agent (ICA) serves as a central coordinator that aggregates trust scores from all Thinker Agents and generates final trust assessments and incentive policies for the entire UAV network. The ICA employs a hierarchical Bayesian inference mechanism to fuse multi-source trust information and produce globally consistent decisions. 459

Global Trust Score Aggregation: The ICA computes the global trust score $\mathcal{T}_k(t)$ for each UAV node k by combining the individual trust scores from all Thinker Agents that have observed or interacted with node k : 460

$$\mathcal{T}_k(t) = \frac{\sum_{j=1}^M \phi_{jk}(t) \cdot \hat{T}_{jk}(t)}{\sum_{j=1}^M \phi_{jk}(t)} \quad (19) \quad 461$$

where $\phi_{jk}(t) \in [0, 1]$ is the confidence weight assigned to Thinker Agent j 's assessment of node k , computed based on the agent's historical accuracy: 462

$$\phi_{jk}(t) = \frac{\exp(-\gamma \cdot \varepsilon_{jk}(t))}{\sum_{j'=1}^M \exp(-\gamma \cdot \varepsilon_{j'k}(t))} \quad (20) \quad 463$$

with $\varepsilon_{jk}(t) = \frac{1}{t} \sum_{\tau=1}^t |\hat{T}_{jk}(\tau) - T_{jk}^{obs}(\tau)|$ representing the mean absolute error of agent j for node k over time, and γ a scaling parameter. 464

Uncertainty Quantification: The ICA also quantifies the uncertainty in global trust assessments using a Bayesian posterior variance: 465

$$\sigma_k^2(t) = \frac{\sum_{j=1}^M \phi_{jk}(t) \left(\hat{T}_{jk}(t) - \mathcal{T}_k(t) \right)^2}{\sum_{j=1}^M \phi_{jk}(t)} + \sigma_0^2 \quad (21) \quad 466$$

where σ_0^2 represents the inherent measurement noise variance. This uncertainty metric is crucial for risk-aware decision-making in zero-trust environments. 467

Final Trust Classification: Based on the global trust score and uncertainty, the ICA classifies nodes into trust categories: 468

$$C_k(t) = \begin{cases} \text{Trusted} & \text{if } \mathcal{T}_k(t) \geq \theta_{high} \text{ and } \sigma_k(t) \leq \sigma_{th} \\ \text{Suspicious} & \text{if } \theta_{low} \leq \mathcal{T}_k(t) < \theta_{high} \text{ or } \sigma_k(t) > \sigma_{th} \\ \text{Malicious} & \text{if } \mathcal{T}_k(t) < \theta_{low} \text{ and } \sigma_k(t) \leq \sigma_{th} \end{cases} \quad (22) \quad 469$$

where θ_{high} and θ_{low} are trust thresholds, and σ_{th} is the uncertainty threshold. 470

The ICA generates global incentive policies that coordinate the actions of individual Thinker Agents to promote network-wide trustworthiness. The incentive policy $\Pi(t)$ is a mapping from global trust assessments to resource allocation decisions: 471

$$\Pi(t) = \left\{ \pi_{jk}(t) : j \in [1, M], k \in \mathcal{N}_j \right\} \quad (23) \quad 472$$

where $\pi_{jk}(t)$ represents the incentive action for node k by agent j , determined through a constrained optimization problem:

$$\max_{\Pi(t)} \mathbb{E} \left[\sum_{k=1}^N U_k(\mathcal{T}_k(t), \pi_{\cdot k}(t)) \right] \quad (24)$$

subject to:

$$\sum_{j=1}^M \sum_{k \in \mathcal{N}_j} c(\pi_{jk}(t)) \leq B_{total}(t) \quad (25)$$

$$\pi_{jk}(t) \in \mathcal{A}_{inc}, \forall j, k \quad (26)$$

where $U_k(\cdot)$ is the utility function for node k , $c(\pi)$ is the cost of incentive action π , $B_{total}(t)$ is the total available resource budget, and $\mathcal{A}_{inc}$ is the set of feasible incentive actions.

The utility function is designed to encourage trust improvement:

$$U_k(\mathcal{T}_k(t), \pi_{\cdot k}(t)) = \mathcal{T}_k(t) \cdot (1 + \delta \cdot \mathbb{I}(\pi_{\cdot k}(t) > 0)) - \kappa \cdot \mathbb{I}(\mathcal{T}_k(t) < \theta_{low}) \quad (27)$$

This formulation rewards high-trust nodes with positive incentives and penalizes low-trust nodes, creating a natural pressure for nodes to maintain trustworthy behavior.

3.3.4. State Update Mechanism for Trust Evolution

The state update mechanism governs how trust scores evolve over time as a function of node behavior, incentive actions, and network conditions. Trust evaluation uses Bayesian inference, and the overall trust dynamics constitute a Markov process because the trust score at time $t + 1$ depends only on the trust score at time t and the current observations and incentives, not on the full history. The state transition can be decomposed as:

$$\mathbf{T}(t+1) = f(\mathbf{T}(t), \mathbf{A}^{inc}(t), \mathbf{A}^{task}(t), \mathbf{E}(t)) \quad (28)$$

where $\mathbf{T}(t) \in \mathbb{R}^N$ is the vector of global trust scores, $\mathbf{A}^{inc}(t)$ is the incentive action matrix, $\mathbf{A}^{task}(t)$ captures task execution outcomes, and $\mathbf{E}(t)$ represents environmental factors. The trust score for each node k evolves according to the following three-stage process:

Temporal Trust Decay: To prevent trust stagnation and enforce continuous verification, we incorporate a temporal decay factor:

$$\tilde{\mathcal{T}}_k(t+1) = \mathcal{T}_k(t) \cdot \exp(-\rho \cdot \Delta t) \quad (29)$$

where $\rho > 0$ is the decay rate, ensuring that trust gradually diminishes without positive reinforcement.

Trust Update Based on Observations: The final trust score is updated via a Bayesian filter that combines the decayed prior with new observations:

$$\mathcal{T}_k(t+1) = \frac{\sigma_k^2(t)}{\sigma_k^2(t) + \sigma_{obs}^2} \cdot \tilde{\mathcal{T}}_k(t+1) + \frac{\sigma_{obs}^2}{\sigma_k^2(t) + \sigma_{obs}^2} \cdot T_k^{obs}(t+1) \quad (30)$$

where σ_{obs}^2 is the observation noise variance, and $T_k^{obs}(t+1)$ is the aggregated observed trust score from all Thinker Agents:

$$T_k^{obs}(t+1) = \frac{1}{|\mathcal{M}_k(t+1)|} \sum_{j \in \mathcal{M}_k(t+1)} T_{jk}^{obs}(t+1) \quad (31)$$

with $\mathcal{M}_k(t+1)$ denoting the set of agents that observed node k at time $t + 1$.

Convergence Analysis: The trust update mechanism in (12)–(14) admits bounded updates under the $[0, 10]$ trust scale. Specifically, since $\mathcal{T}_k(t) \in [0, 10]$ and $\tilde{\mathcal{T}}_k(t+1) = \mathcal{T}_k(t) \cdot \exp(-\rho\Delta t) \in [0, 10]$, and the Bayesian fusion in (14) is a convex combination of two bounded terms, the trust score remains in $[0, 10]$ for all t . This boundedness ensures that the update process does not diverge.

For convergence, we note that the temporal decay factor $\exp(-\rho\Delta t) < 1$ for $\Delta t > 0$ contracts the prior toward zero, while the observation term $T_k^{obs}(t+1) \in [0, 10]$ is bounded. The resulting process is a stochastic approximation with a contracting step size, which under standard conditions (bounded increments, martingale convergence theorem) converges almost surely to a fixed point [37]. We leave rigorous convergence proofs for the full DecPOMDP to future work, as the decentralized partial observability introduces coupling terms that require additional technical machinery (e.g., [38]). Empirically, Section 4 demonstrates that trust scores converge within the 5000 s simulation window for all configurations tested.

Trust-Driven Network Evolution: The state update mechanism creates a positive feedback loop where trustworthy behavior is rewarded with higher trust scores and corresponding incentives, while malicious behavior leads to trust degradation and restricted access. Over time, this mechanism induces a convergence effect where the distribution of trust scores across the swarm stabilizes, as characterized by the mixing bound in (21). Empirically, the simulation results in Section 4 confirm that trust scores converge within the 5000 s mission window.

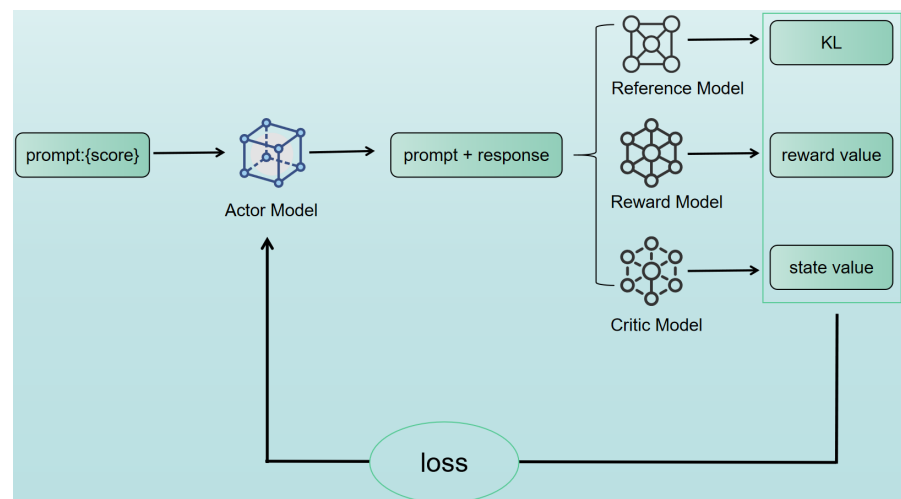


Figure 2. State update mechanism in MTIM: trust scores evolve through a Bayesian update process combining temporal decay, multi-agent observations, and incentive-driven feedback.

3.3.5. Evaluation Metrics

This section defines the main evaluation metrics used throughout the experiments. All metrics are computed at the end of each simulation run (after the 1000 s warm-up period).

Trust Classification Metrics: For malicious-node detection, let TP (true positive) denote correctly identified malicious nodes, TN (true negative) correctly identified benign nodes, FP (false positive) benign nodes misclassified as malicious, and FN (false negative) malicious nodes misclassified as benign. The detection accuracy is:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \quad (32)$$

The precision and recall are:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \quad (33)$$

The F-score is:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (34)$$

Ground-truth labels are determined by the predefined attack pattern assignment at simulation initialization; nodes assigned as malicious in the simulation ground truth are treated as ground-truth malicious throughout.

Mission-Level Metrics: The task success rate (TSR) measures the fraction of successfully completed tasks among all initiated tasks:

$$\text{TSR} = \frac{N_{\text{succ}}}{N_{\text{all}}}, \quad (35)$$

where a task is deemed *successful* if it completes before mission timeout with at least one trusted UAV contributing, and *failed* if it times out or is assigned exclusively to Suspicious/Malicious UAVs. TSR is reported separately from detection metrics to distinguish trust evaluation quality from task allocation performance.

Communication Overhead: The average bytes transmitted per UAV per trust update cycle, encompassing trust sub-score broadcasts, Thinker-to-ICA messages, and incentive signals.

4. Performance Evaluation

4.1. Simulation Setup

Experiments were conducted on a workstation equipped with four NVIDIA HGX H20 GPUs (96 GB each) running Ubuntu 20.04 LTS. The simulation and learning modules were implemented in Python and PyTorch, while network interactions were modeled in ns-3 [39]. The trust evaluation pipeline uses a lightweight Q-network (3-layer MLP with hidden sizes 128-256-128) for Q-value estimation; the Qwen2.5-7B model [36] is used solely for generating natural-language trust reports and explaining incentive decisions to human operators in the ground control station interface, and does not participate in the real-time trust scoring loop. This human-in-the-loop component runs asynchronously with a latency tolerance of seconds, far exceeding the 10-second trust update interval. As summarized in Table 1, the parameter settings were selected to approximate realistic low-altitude urban UAV scenarios.

The simulation area was set to 10 km × 10 km with a duration of 5000 s to capture sufficient trust evolution dynamics. The first 1000 s were used as a warm-up period during which no trust decisions were enforced, allowing agents to accumulate initial observations. UAV mobility follows a Random Waypoint model with pause time of 5 s and maximum speed of 15 m/s, re-sampling waypoints uniformly within the simulation area. Malicious UAVs adopt three representative attack patterns: (1) *constant attack* where malicious nodes exhibit low trust from the start and maintain adversarial behavior throughout; (2) *dormant attack* where nodes appear benign initially but switch to malicious behavior after warm-up; and (3) *on-off attack* where malicious behavior toggles every 500 s to simulate intermittent adversaries. Each attack pattern is evaluated separately and accounts for one-third of the malicious nodes in mixed-attack experiments.

To evaluate scalability, the number of UAVs varied from 5 to 20, and each configuration was repeated across 50 independent trials with different random seeds. Initial energy ranged from 50 to 200 Wh, operational power consumption from 100 to 120 W, and UAV

Table 1. Simulation parameters tailored to the proposed multi-agent trust evaluation architecture.

Parameter	Value
Simulation area	10 km × 10 km
Simulation time	5000 s
Warm-up period	1000 s
Number of UAVs	5 – 20
Number of trials	50
Initial energy	50–200 Wh
Power consumption	100 – 120 Watts
UAV speed distribution	3–15 m/s
Inter-agent communication bandwidth	5 Mbit/s
UAV communication range	300 m
Path loss coefficient	2.5 – 4 dB
Transmission power	0.1 – 1 W
Number of Thinker Agents	5
Number of Inferential Conclusion Agents	1
Trust factor update interval	10 s
Incentive policy update interval	50 s
Temporal trust decay rate ρ	0.01
Trust threshold θ_{high}	7.5
Trust threshold θ_{low}	3.0
Uncertainty threshold σ_{th}	0.5

speed from 3 to 15 m/s. Inter-UAV bandwidth was fixed at 5 Mbit/s, communication range at 300 m, path-loss coefficient at 2.5–4, and transmission power at 0.1–1 W. The trust engine used 5 Thinker Agents and 1 Inferential Conclusion Agent. Trust-factor updates were performed every 10 s, incentive updates every 50 s, and the temporal decay rate was set to $\rho = 0.01$. Trust classification used $\theta_{\text{high}} = 7.5$, $\theta_{\text{low}} = 3.0$, and $\sigma_{\text{th}} = 0.5$. All results are reported as mean $\pm$ standard deviation across trials.

Communication overhead is measured as the average bytes exchanged per UAV per trust update cycle (10 s interval), including trust sub-score broadcasts, agent-to-ICA messaging, and incentive signal distribution. Energy consumption is tracked continuously and reported as average Watt-hours consumed per UAV over the mission duration.

To better position MTIM against representative baselines, four widely cited trust management schemes were re-implemented under the same experimental setting:

- **FUBA [8]:** A fuzzy-based behavior analytics scheme for FANET trust evaluation. It integrates heterogeneous indicators (RSSI, PDR, delay, battery level) through fuzzy logic rules to compute node credibility scores. The original fuzzy rule set was implemented with five trust dimensions as fuzzy inputs.
- **GALTrust [10]:** A generative adversarial learning-based trust framework that adapts to emerging malicious behaviors. It uses adversarial training to detect previously unseen attack patterns. Its anomaly scoring mechanism was adapted to the multi-factor trust dimensions.
- **GDM-DTM [23]:** A group decision-making dynamic trust management scheme that combines multi-dimensional trust aggregation with manual weights. Its trust fusion was re-implemented with the manual weight configuration from the original paper.
- **Multi-factor Average (MFA):** A baseline that computes trust as the equal-weighted average of five trust dimensions without learning or adaptation. This serves as a naive fusion reference to isolate the contribution of MARL-based adaptation.

All baselines were implemented in PyTorch using the same network topology, UAV mobility model, and observation streams as MTIM to ensure a strictly controlled comparison. Each baseline was tuned to its original parameter settings within the framework.

Table 2. Performance comparison of MTIM with baseline methods under the unified experimental setting (30% malicious-node ratio, 15 UAVs, 5000 s simulation). All values are reported as mean $\pm$ standard deviation over 50 independent trials.

Method	Accuracy	F-score	TSR	Comm. Overhead (KB/UAV)	Eval. Time (ms)
MFA	64.8% $\pm$ 6.1%	59.2% $\pm$ 6.8%	61.4% $\pm$ 5.2%	12.3 $\pm$ 2.1	8.2 $\pm$ 1.5
FUBA [8]	73.5% $\pm$ 5.4%	69.8% $\pm$ 5.9%	72.1% $\pm$ 4.8%	18.7 $\pm$ 3.2	15.4 $\pm$ 2.8
GALTrust [10]	81.2% $\pm$ 4.6%	77.6% $\pm$ 5.1%	78.9% $\pm$ 4.3% [†]	24.1 $\pm$ 4.5	38.2 $\pm$ 6.7
GDM-DTM [23]	82.4% $\pm$ 4.2%	80.1% $\pm$ 4.8%	82.1% $\pm$ 3.8% [†]	15.8 $\pm$ 2.9	18.5 $\pm$ 3.4
MTIM (ours)	93.0% $\pm$ 3.2%	89.4% $\pm$ 4.1%	90.0% $\pm$ 3.1%[†]	28.6 $\pm$ 5.1	52.4 $\pm$ 9.6

(*) MFA = Multi-factor Average (equal weights, no learning). (†) All methods use 5 trust dimensions for fair comparison. (‡) TSR = Task Success Rate; Comm. Overhead = per-UAV communication overhead per update cycle. (‡) Eval. Time = trust evaluation latency per update interval. (b) Improvements over best baseline (GDM-DTM) are statistically significant (paired t-test, $p < 0.01$) for all metrics. (*)‡ TSR values for GALTrust, GDM-DTM, and MTIM include the effect of incentive mechanisms on task allocation.

The results in Table 2 demonstrate that MTIM achieves the highest accuracy (93.0%), F-score (89.4%), and task success rate (90.0%) among all compared methods, with statistically significant improvements (paired t-test, $p < 0.01$) over all baselines. While MTIM incurs higher evaluation overhead due to the MARL training and Bayesian inference, this cost is amortized over the mission lifetime and remains within real-time constraints for the 10-second trust update interval specified in Table 1. The three practical advantages of MTIM are: (1) continuous trust re-evaluation from heterogeneous observations during network operation; (2) explicit modeling of aggregation uncertainty, which is critical when local observations are noisy or sparse; and (3) MARL-based adaptation that replaces manually designed adjustment rules, allowing trust-factor weighting and incentive policies to evolve with network conditions.

4.2. Trust Evolution Analysis

Figure 3 shows the temporal evolution of the average trust scores of benign and malicious UAVs under different numbers of agents (2, 5, and 10). The simulation spans 5000 s with the first 1000 s as warm-up, illustrating how trust values evolve under the proposed framework.

The results show a clear separation trend between benign and malicious nodes across all agent configurations. Benign UAVs exhibit gradually increasing trust, whereas malicious UAVs experience sustained trust degradation. Increasing the number of agents slows convergence slightly because more local assessments must be fused, but it also stabilizes the global estimate by reducing the influence of any single local observer.

This behavior is consistent with the design of the proposed framework. Parallel local assessment broadens observation coverage, Bayesian aggregation suppresses isolated estimation errors, reward-driven policy adaptation improves factor weighting over time, and temporal decay prevents stale trust from masking recent malicious behavior.

4.3. Comparative Performance Analysis

Figure 4(a) presents the anti-manipulation performance of the proposed framework across varying malicious-node ratios, comparing the detection accuracy of the five individual trust branches with the aggregated trust score generated by the Inferential Conclusion

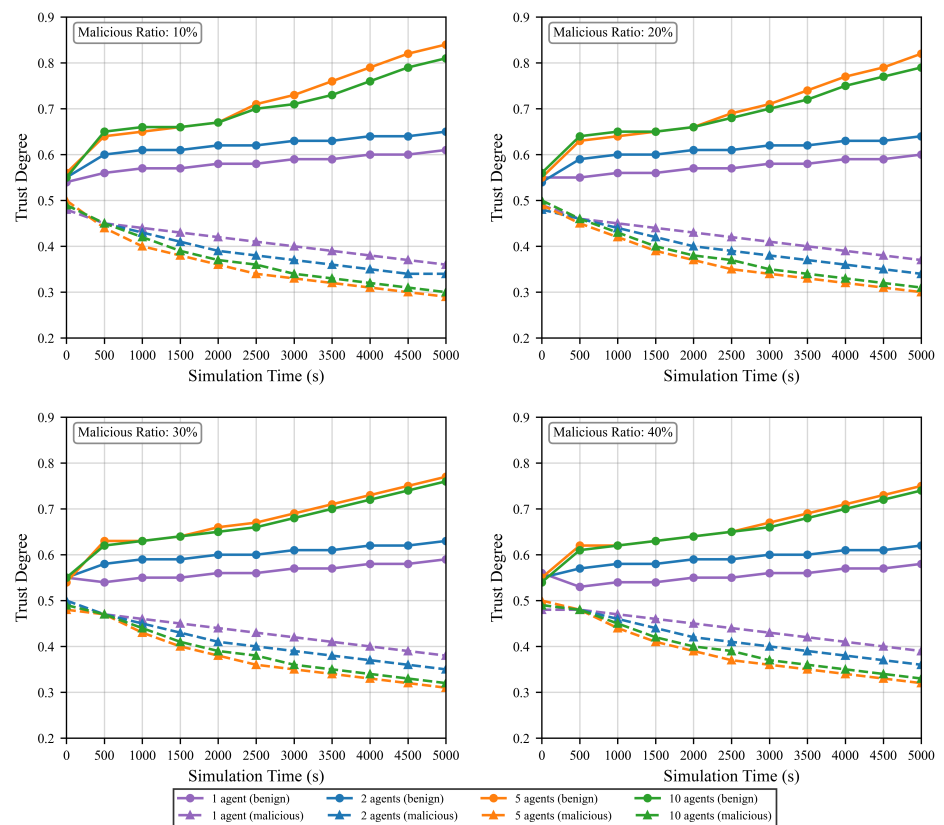


Figure 3. Trust evolution of benign and malicious UAVs under different numbers of agents.

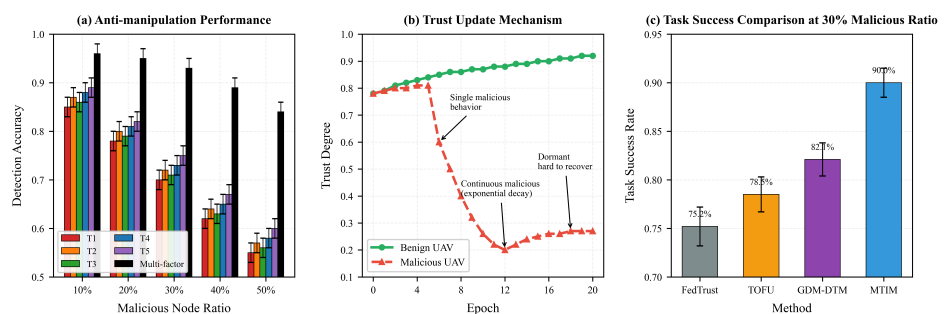


Figure 4. Performance comparison results: (a) detection accuracy versus malicious-node ratio; (b) trust evolution of benign and malicious UAVs across epochs; (c) task success comparison of FUBA, GALTrust, GDM-DTM, and MTIM at a 30% malicious-node ratio.

Agent (ICA). The five trust branches correspond to behavioral trust (T1), communication trust (T2), energy trust (T3), task-execution trust (T4), and historical trust (T5).

Figure 4(b) evaluates the trust update mechanism under three malicious-behavior patterns. Benign nodes maintain stable trust, continuously malicious nodes experience rapid trust degradation, and dormant attackers do not regain trust immediately after inactive periods. This result shows that the update rule can resist both persistent and intermittent attacks, which is consistent with the zero-trust requirement for continuous verification.

Figure 4(c) compares four representative methods at a malicious-node ratio of 30%. MTIM achieves the highest task success rate at 90.0%, followed by GDM-DTM at 82.1%, GALTrust at 78.9%, and MFA at 61.4%. This ranking is consistent with Table 2 and indicates that the proposed framework better preserves mission continuity under adversarial participation. The gain can be attributed to continuous multi-factor trust re-evaluation and

adaptive incentive adjustment, which together reduce the impact of unreliable nodes on collaborative task execution.

Taken together, the three subfigures support the main claims of the proposed framework: multi-factor aggregation is more resistant to manipulation than any single trust branch, dynamic trust updates maintain trust separation under changing attack patterns, and RL-based incentives improve behavioral differentiation relative to static alternatives.

4.4. Ablation Study

To verify the contribution of each component in MTIM, we conduct an ablation study by progressively removing or replacing key mechanisms. The baseline is the full MTIM with all components enabled. Each ablation variant is evaluated under the same unified setting with 30% malicious nodes and 15 UAVs over 50 trials.

Table 3. Ablation study of MTIM components (30% malicious-node ratio, 15 UAVs). All values are mean +/- std over 50 trials.

Variant	Accuracy	F-score	TSR
MTIM (full)	93.0% ± 3.2%	89.4% ± 4.1%	90.0% ± 3.1%
w/o Cross-factor Checking	86.4% ± 4.8%	82.1% ± 5.2%	83.7% ± 4.2%
w/o Bayesian Aggregation	84.2% ± 5.1%	79.8% ± 5.6%	81.3% ± 4.6%
w/o Temporal Decay	88.7% ± 4.0%	84.9% ± 4.7%	86.2% ± 3.8%
w/o MARL Adaptation	79.3% ± 5.8%	74.6% ± 6.1%	76.8% ± 5.2%
w/o Incentive Mechanism	77.8% ± 6.0%	72.4% ± 6.4%	74.5% ± 5.5%
T1 Only	64.2% ± 7.2%	58.7% ± 7.8%	62.3% ± 6.8%
T2 Only	61.8% ± 7.5%	55.4% ± 8.1%	59.6% ± 7.1%
T3 Only	58.4% ± 8.0%	52.1% ± 8.5%	56.2% ± 7.6%

The ablation results in Table 3 demonstrate that each component contributes meaningfully to MTIM’s overall performance. Removing the MARL adaptation mechanism causes the largest drop (13.7 percentage points in accuracy), confirming that adaptive trust-factor weighting is the core innovation. The incentive mechanism also contributes substantially (15.2 pp drop without incentives), validating the closed-loop trust-incentive design. Removing ICA (Bayesian aggregation) reduces accuracy by 8.8 pp, showing the importance of multi-agent fusion. Cross-factor consistency checking provides a 6.6 pp improvement by detecting coordinated attacks. Temporal decay contributes 4.3 pp by preventing stale trust. Finally, single trust factor variants perform significantly worse (26–35 pp below full MTIM), justifying the multi-factor approach.

5. Conclusions

This paper presented MTIM, a MARL-based trust evaluation framework for zero-trust low-altitude UAV networks. MTIM integrates five trust dimensions, cross-factor consistency checking, decentralized trust learning, Bayesian global aggregation, and temporally decayed trust updates. Compared with representative baselines, MTIM improves malicious-node screening and mission-level task continuity under adversarial conditions, demonstrating that adaptive multi-factor trust learning is more effective than static or single-branch trust assessment in dynamic UAV environments.

The main implication of this study is that trust management in low-altitude UAV networks should be formulated as a sequential decision problem rather than a fixed scoring task. By continuously updating trust from heterogeneous evidence and adapting incentives online, MTIM better matches the operational requirements of zero-trust UAV systems. Future work will focus on reducing communication overhead, extending the method to cross-domain missions, and integrating secure distributed storage to improve trust portability and auditability.

Author Contributions: Author contributions will be finalized after confirmation of the final author list and roles.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

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